THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 10: **The Human Eye and the Colourful World** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The ability of the eye lens to adjust its focal length is called: 1

(a) accommodation **(b)** dispersion
(c) persistence of vision **(d)** power of lens

2. The least distance of distinct vision for a young adult with normal vision is: 1

(a) 25 cm **(b)** 25 m **(c)** infinity **(d)** 2.5 cm

3. Myopia can be corrected by using a: 1

(a) cylindrical lens **(b)** concave lens **(c)** convex lens **(d)** bifocal lens

4. A student states: “The sky appears blue because of scattering of light.” The correct response is: 1

(a) Correct **(b)** Incorrect — due to dispersion
(c) Incorrect — due to reflection from sea **(d)** Cannot be determined

5. For the colour of light that deviates the least through a prism, Student A says red; Student B says violet. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. The splitting of white light into its component colours is called: 1

(a) dispersion **(b)** reflection **(c)** scattering **(d)** refraction

7. Twinkling of stars is due to: 1

(a) dispersion **(b)** scattering
(c) atmospheric refraction **(d)** total reflection

8. The coloured diaphragm that controls the size of the pupil is the: 1

(a) retina **(b)** ciliary muscle **(c)** iris **(d)** cornea

9. The danger signal lights are red because red light: 1

(a) travels fastest **(b)** is the brightest
(c) is scattered the least by fog/smoke **(d)** is scattered the most

10. The defect in which a person cannot see nearby objects clearly is: 1

(a) myopia **(b)** hypermetropia **(c)** presbyopia only **(d)** cataract

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the far point and near point of a normal human eye? 2

12. Why do stars twinkle while planets do not? 2

13. What is presbyopia? 2

14. A student says that the sky would appear black to an astronaut in space. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is myopia? State its two causes. How is it corrected? 3

16. Explain why the Sun appears reddish at sunrise and sunset. 3

17. Draw (describe) the path of a ray of light through a glass prism. Label the angle of deviation. 3

OR

What is a rainbow? How is it formed? 3

18. What is hypermetropia? Give two causes and its correction. 3

OR

Explain advanced sunrise and delayed sunset. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the structure of the human eye with a labelled diagram (main parts). 5
(ii) Explain the functions of retina, cornea, ciliary muscles and optic nerve.

OR

(i) What is dispersion? 5
(ii) Why do different colours deviate differently?
(iii) Describe Newton's experiment with two prisms.
(iv) What is a spectrum?

20. **Case study:** A student sitting at the back of a class cannot read the board clearly but can read a book easily. 5
(i) Name the defect.
(ii) Where is the image formed in his eye?
(iii) Name the lens used to correct it.
(iv) State two causes of the defect.

OR

(i) What is scattering of light? 5
(ii) Explain the Tyndall effect with an example.
(iii) Why is the sky blue?
(iv) Why does the sky look white on a clear day at noon with large particles?

END OF QUESTION PAPER — Total: 40 Marks